AI for Education

CA1 Progress Report

Document status
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Implementation
One student

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Project	AI-Powered Multimodal Content & Assessment Generation for Educational Applications
Supervisor	Mehul Motani
Period	AY 2026/2027 October 2026 to May 2027
Focus	CA1 Literature review and course-grounded text generation

1 Abstract and CA1 scope

This report presents the literature review, system design and initial implementation of an API-based educational content and assessment generator. The system is intended to turn course material into explanations, quizzes and assignments with explicit difficulty constraints and traceable source references. Text-grounded generation is the CA1 focus; speech and visual outputs remain part of the full project.

The literature review identifies three distinctions that shape the design: factual correctness is not guaranteed by retrieval, cognitive demand is not identical to observed student difficulty, and a valid citation identifier does not establish that a source supports a claim. The proposed evaluation therefore separates software validity, evidence support and pedagogical suitability.

A local web prototype implements document parsing, vector retrieval, structured generation, editing, review and evidence export. Nine automated integration tests pass using isolated HTTP fixtures. A previous browser check demonstrated course creation and text import. Real text and embedding providers have not yet been configured, so the core live RAG demonstration remains outstanding. No model-quality or learning-effectiveness results are claimed.

CA1 requirement [1]	Material prepared	Current readiness
Literature survey	Critical review and 14 research references	Working manuscript ready for review
System design	Architecture, data flow, API and version rules	Documented and partly validated in code
Initial functional text plus RAG prototype	Local application, tests and live smoke procedure	Implementation present; real API demonstration pending

The supplied brief does not specify a report length, report template or CA1 date. This document is therefore organised as an English progress report for supervisor discussion. It must not be represented as a completed live prototype submission until the outstanding provider test and manual content review have been performed.

2 Review scope and organising questions

This review examines how an educational authoring system can produce course-grounded explanations and assessment items while making its limitations inspectable. It supports the CA1 literature-survey and system-design requirements of the project brief [1]. The organising question is not simply whether a language model can generate a fluent question, but what evidence is needed before an educator should use that question.

A targeted narrative search was conducted on 11 September 2026 using ACL Anthology, arXiv and PMLR. Search themes included educational question generation, difficulty estimation, retrieval-augmented generation, citation evaluation and multimodal RAG. Fourteen research publications from 2017–2025 were selected for methodological relevance. This is not an exhaustive systematic review or a meta-analysis, and no claim is made that all relevant 2026 publications have been covered.

Official publication pages and downloaded full texts were used to check the selected methods, evaluation designs and limitations. Peer-reviewed conference or journal publications form the main evidence base; two arXiv preprints are explicitly distinguished. Publication records take precedence over placeholder venue information in preprint PDFs. The accompanying source ledger records the inspected sections and local copies so that important claims can be checked again.

The synthesis separates four issues: pedagogical suitability, retrieval of supporting material, interpretation of difficulty, and evaluation of generated outputs. Multimodal presentation is reviewed as the next development stage, while the implemented CA1 path remains text-based. Subsequent paragraphs distinguish reported findings from design implications proposed for this project.

A common comparison problem is that papers use different subjects, data, models and evaluation procedures. Their results therefore motivate hypotheses rather than directly predicting performance on the selected FYP course. The report does not combine their benchmark scores into a ranking. Instead, it asks which assumptions transfer to a small, API-based, educator-reviewed system and which require new evidence.

3 Educational question generation

Du, Shao and Cardie [2] formulate reading-comprehension question generation with an attention-based sequence-to-sequence model. Their sentence-versus-paragraph comparison and human evaluation show that neural generation can move beyond hand-crafted rules. The study concerns passage-based questions, however, rather than a complete course-authoring workflow.

Gong, Pan and Hu [3] introduce KHANQ, built from learner questions in science-related Khan Academy courses. Their evaluation considers appropriateness, coverage, coherence and complexity. The work exposes a useful distinction between producing a question that can be answered from text and producing one that probes understanding.

Maity, Deroy and Sarkar [4] compare in-context learning, retrieval-based generation and a hybrid approach for educational questions. The preprint uses GPT-4 for ICL and BART with retrieval for its RAG setting. Consequently, differences between these systems cannot automatically be attributed to retrieval alone.

Implications for the proposed system

For this project, the authoring unit is a learning asset containing an objective, an explanation, questions, reference answers and learner-facing explanations. This is an explicit design choice: an isolated question provides insufficient information for checking whether it tests the intended idea or whether the proposed answer follows from the course. The teacher must be able to inspect these elements together.

The comparison also motivates a controlled baseline. An API-only model without retrieved course passages should be compared with the same model receiving those passages. The prompt, question count, requested language and output format should otherwise remain comparable. Introducing a stronger model in only one condition would confound the contribution of retrieval with the contribution of the generator.

Multiple-choice and short-answer questions are the initial formats. Multiple-choice items need checks for a unique defensible answer and plausible distractors; short-answer items need an explicit marking expectation. Four distinct strings do not guarantee four pedagogically useful options. These judgments remain separate from the implemented structural validator and require inspection by someone familiar with the course.

The resulting research focus is therefore narrower than autonomous curriculum creation: generate reviewable drafts within a specified course, record the evidence used, and measure the amount and type of correction needed before use. This definition provides observable outcomes without presuming that generation automatically saves educator time.

4 Difficulty and educational objectives

Cheng et al. [5] propose EQPR, combining planning and reflection for objective-aligned mathematical question generation, with EduMath and EQGEVAL supporting evaluation. Their approach illustrates that educational objectives can be represented and assessed explicitly. Its mathematics-specific planning machinery is not evidence that the same procedure transfers unchanged to every course.

Park et al. [6] use groups of language models to estimate question difficulty for different learner levels. The paper evaluates LLaSA and a zero-shot variant against student-derived difficulty information. Its findings concern difficulty estimation; they do not establish that a generator obeys a requested difficulty label.

Intended demand versus observed difficulty

The present design treats easy, medium and hard as authoring constraints on intended cognitive demand. An easy item targets a single concept; a medium item applies or compares concepts in a stated situation; a hard item requires integration or justification within the available course content. These are provisional operational definitions, not a validated psychometric scale.

A question can require several reasoning steps yet be easy for an experienced student. Conversely, unfamiliar terminology or unclear wording can make a simple concept difficult. The experiment must therefore avoid interpreting a teacher's complexity rating as a measured student success probability. Actual difficulty would require an appropriate learner population, response observations and a justified analysis.

The initial study will compare a difficulty label alone with a label plus explicit cognitive constraints, keeping the generator and retrieval settings fixed. Raters should assign their perceived demand category before seeing the requested label. A second comparison can then measure intended-versus-rated agreement. Ambiguous, incorrect or out-of-scope questions are recorded as defects rather than rewarded for appearing difficult.

The prototype currently asks for a brief difficulty rationale, which helps a teacher understand the intended demand. That rationale is generated by the same model and is not independent verification. A model claiming that a question is hard must not be counted as successful difficulty control. The CA1 contribution is the explicit design and testable comparison; calibration to real learners remains future work.

5 Retrieval and evidence grounding

Lewis et al. [7] combine a parametric generator with retrieved non-parametric memory and evaluate jointly trained RAG formulations. Their work motivates explicit external knowledge access. The present prototype instead uses inference-time prompting through external APIs and does not reproduce their end-to-end training procedure.

Karpukhin et al. [8] develop dense passage retrieval using a dual encoder for open-domain question answering. This provides a foundation for semantic passage selection. Its benchmark results do not establish that a chosen embedding API retrieves course-specific terminology better than a lexical baseline.

Liu et al. [9] find that model performance can depend on the position of relevant information in long contexts. Their controlled tasks caution against equating context-window capacity with reliable context use. The effect should be rechecked for the eventual provider rather than assumed constant across models.

A BOUNDED RETRIEVAL DESIGN

The CA1 implementation splits text within each PDF page, preserving document identity and page number. It embeds these chunks and retrieves the five most similar chunks for a requested learning objective. Cosine similarity ranks candidates; it is not interpreted as the probability that a passage is correct. The selected chunks are saved with the generated asset, making later inspection independent of changes to the original index.

This simple design is a starting point with clear limitations. Page boundaries can split an explanation, character-based chunks may contain several ideas, and a fixed top-five selection can return irrelevant material for unsupported topics. There is no validated relevance threshold or reranker in the current implementation. An insufficient-evidence response is requested from the generator, but reliable abstention still needs testing.

Three failure types must be distinguished: missing support in the uploaded collection, failure to retrieve support that is present, and failure to use retrieved support correctly. They require different fixes. A retrieval error should not be hidden by improving the answer prompt, while a hallucinated answer should not be explained away merely because a relevant passage was available.

Index configuration is also part of the experimental state. The system records an embedding configuration signature and prevents mixed use after a change. This guards against a concrete engineering inconsistency, although it cannot detect an undocumented upstream model revision. Later retrieval experiments should include lexical retrieval, alternative chunk sizes and annotated support passages before adding more complex infrastructure.

6 Evaluation of questions and citations

Gao et al. [10] introduce ALCE and distinguish output correctness from citation quality. Citation recall and precision examine support and irrelevant citations. This distinction informs the project's separation between valid source identifiers and substantive evidence for a claim.

Es et al. [11] propose RAGAs metrics for faithfulness, answer relevance and context relevance using model-based judgments. Saad-Falcon et al. [12] propose ARES, which trains evaluators on synthetic data and uses human annotations for prediction-powered inference. These approaches differ in supervision and calibration requirements.

Oh et al. [13] show why single-reference similarity is restrictive for question evaluation and propose multiple reference paraphrases. Their stated limitations include dependence on an existing reference and restrictions for answer-unconditional generation. Multiple valid questions can therefore make a single reference misleading.

Deroy, Maity and Sarkar [14] propose MIRROR, using iterative model feedback for open-ended question evaluation. Their reported closer alignment with human ratings motivates investigating assisted review, while leaving the need for an independent reference assessment in this course unresolved.

AN EVALUATION HIERARCHY

The proposed hierarchy begins with deterministic checks: required fields, question count, answer format and source identifiers. These checks establish that an output can be processed, not that it should be taught. The next level is expert assessment of correctness, support, answerability and intended demand. Model-based scoring may then assist triage, provided disagreement with human judgments is measured.

The study will keep generation and judgment logically separate. Using the same model to generate, explain and grade its own output could reproduce the same error three times. Agreement between two model calls is therefore not a substitute for a course reference. If an automated judge is later used, its identity, prompt, calibration subset and failure cases must be retained.

Reporting only accepted outputs would also hide a system that rejects most requests or spends many calls repairing them. Results should include all submitted requests, failure and abstention rates, and conditional quality among valid outputs. Raw drafts and teacher-corrected versions should be evaluated separately so that manual improvements are not credited to the generator.

For CA1, a small technical smoke test demonstrates data flow and catches integration failures. It is not a sufficiently powered study of educational quality. The later research dataset will contain held-out objectives and annotated support evidence; success on the self-authored demonstration material cannot establish generalisation to an unseen university course.

7 Multimodal extension and accessibility

Mei et al. [15] survey multimodal RAG, including retrieval and generation involving different data types. Their taxonomy helps distinguish multimodal evidence processing from presenting text in alternative formats. The current project begins with text retrieval and plans speech and visual transformations after text review.

This distinction matters for the project's claims. Producing an audio narration and an illustration does not mean that the retrieval component understands diagrams in a scanned textbook. The CA1 parser currently extracts text; OCR, image retrieval and visual reasoning over source documents are not implemented. The report therefore uses "multimodal presentation" for the planned output stage.

The proposed CA2 extension binds each media item to the version of the reviewed text that produced it. A teacher checks terminology and omissions in speech, and relationships, labels and unsupported details in images. A revised explanation invalidates the assumption that existing media still represent the material accurately. Separate review states make this dependency visible.

Alternative formats should offer a practical choice, such as listening to an explanation or inspecting a diagram alongside its text. No claim is made that matching an assumed learning style improves learning outcomes. Accessibility must be checked through concrete features and tasks: readable text, playback controls, transcripts, useful alternative descriptions and avoidance of information available only in one modality.

For the CA1 milestone, documenting these interfaces is sufficient preparation for the later multimodal work; a fully integrated multimodal application belongs to CA2 in the supplied brief. The current implementation already contains media adapters and review logic, but neither those interfaces nor a picture prompt constitute evidence that real media outputs have been validated. This boundary keeps the present milestone focused on the text-and-retrieval path.

8 Synthesis and research position

Across the selected literature, generation quality depends on more than linguistic fluency. Question-generation studies identify pedagogical criteria, difficulty work introduces learner or objective dependence, and retrieval evaluation separates finding evidence from using it. The proposed project brings these concerns into a single inspectable authoring workflow, rather than claiming a new foundation model.

The opportunity identified here is a practical and testable integration problem: can a small API-based system make course-grounded question creation more reliable through explicit source links, cognitive constraints and teacher review? The review does not establish that no previous educational system combines these features. A novelty claim would require a broader comparison of systems and published implementations.

Research question	Controlled comparison	Primary outcome
RQ1 Course grounding	Same generator without retrieval versus with course retrieval	Correctness and evidence support
RQ2 Intended demand	RAG with labels versus RAG with explicit cognitive constraints	Blind-rated demand agreement
RQ3 Later media stage	Reviewed text and its generated speech or visual output	Consistency and required corrections

The first two questions guide CA1 design and the subsequent text experiments. RQ3 is reserved for the multimodal stage. Hypotheses are directional expectations to be tested, not established project results. An unsuccessful comparison is still informative if the record explains whether the cause was retrieval, generation, evaluation ambiguity or implementation failure.

The corresponding design commitments are modest but concrete: preserve the source snapshot, version the generated asset, make structural checks explicit, require review before learner display, and record failed attempts. These commitments create the evidence needed to investigate quality. They do not remove the need for real provider testing, course expertise or careful human evaluation.

Before formal submission, the review should be refreshed for newly relevant work and adjusted to the supervisor’s preferred course and rubric. The current manuscript is a complete working narrative for discussion, with a traceable reference set; its scope and methodological decisions remain open to revision as the project gains actual course data.

9 System architecture and requirements

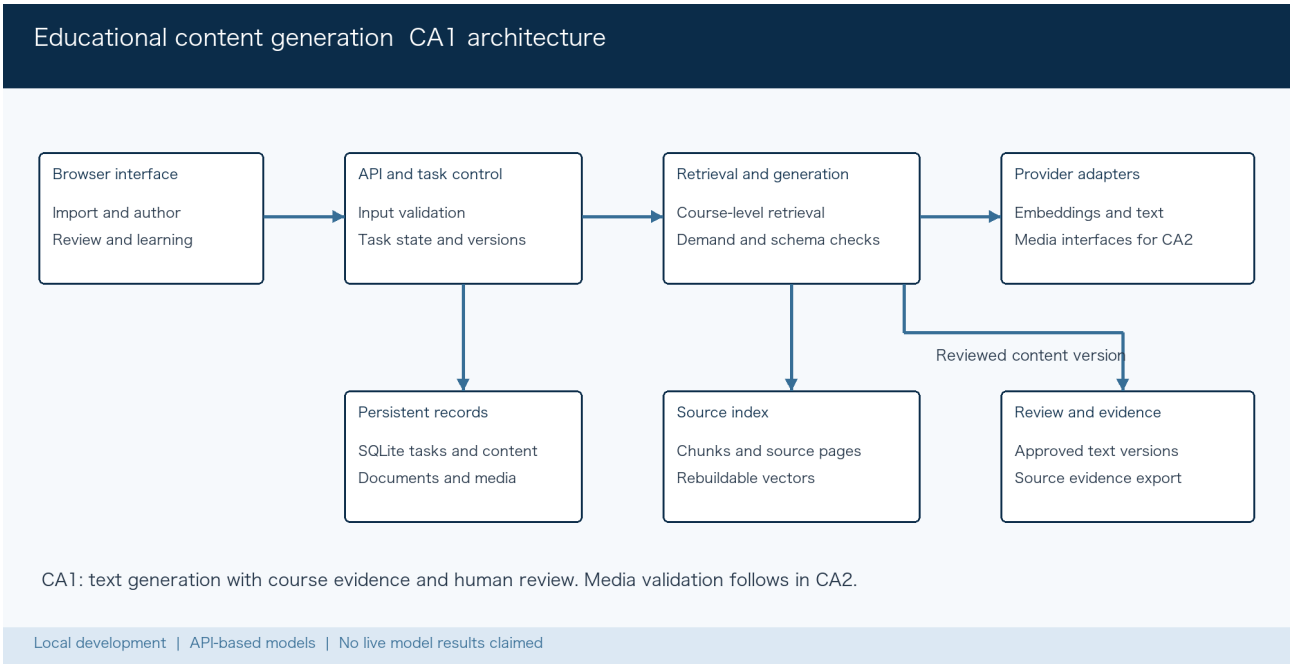


Figure 1 CA1 architecture and extension points

The prototype uses a browser interface, a FastAPI application, a single-process asynchronous task queue, and SQLite with local file storage. Model calls are isolated behind capability adapters. This structure keeps the initial system small enough to inspect while preserving clear replacement points for later providers. It is a local research prototype, not a multi-user deployment.

Component	Responsibility	Boundary
Document and retrieval pipeline	Parse pages, split chunks, embed and rank within a course	Text extraction only; no OCR
Generation and validation	Request structured material and validate fields and references	No automatic proof of factual support
Content and review service	Store source snapshots, revisions and approval state	Teacher judgment remains necessary
Provider adapters	Text, embedding, speech and image HTTP contracts	Real compatibility requires per-provider tests

The CA1 acceptance path is: import a course document, build its index, request a text asset, inspect its evidence, edit or approve it, and open its learning view. Both explanations and assessment items should be demonstrated. Teacher actions and learner display are separate views, but authentication and course authorisation are not yet implemented. The service must remain on the local machine during development.

10 Data flow and reproducibility

During ingestion, a file is associated with a course and identified by a SHA-256 digest. PDF text is extracted page by page; UTF-8 text and Markdown are also accepted. Each page is divided into chunks of at most 1,200 characters, with approximately 120 characters of overlap. These are initial engineering settings, not empirically optimised choices.

Indexing sends chunks to the embedding adapter in batches and stores the vectors with an endpoint-and-model configuration signature. At generation time, the objective is embedded, candidates are restricted to the course, and cosine similarity selects up to five chunks. If the embedding signature has changed, generation is blocked until the index is rebuilt. The retrieved snapshot becomes part of the saved asset.

The generator receives the objective, material type, question type, intended difficulty, language, output schema and retrieved passages. It returns a structured object. Validation checks the expected number of questions, options and answer format, duplicate stems, and whether references belong to the supplied set. A schema or rule failure permits one further generation attempt; upstream transport failures are not silently retried.

Stored record	Essential fields	Reason
Source	Course, document, page, chunk and text	Reconstruct the provided evidence
Generated asset	Text, questions, sources, model and prompt version	Inspect the original generation context
Revision	Content ID, version and edited asset	Separate generated drafts from teacher changes
Task and call	Request key, state, error, model and reported usage	Track failures and repeated work

An identical generation request key and payload reuse the task. A different payload with the same key is rejected. Editing creates a new draft version, requiring review again. The evidence export includes the content, source snapshot, configuration, initial generation call records and revisions; it excludes API credentials. Provider names and signatures improve traceability but cannot guarantee reproducibility when a vendor silently updates a model.

Current limits are 10 MB per file, 300 PDF pages, 400 chunks and 10 questions per request. Daily API-call limits count attempted requests, including failures. These limits are operational safeguards, not an experimentally justified capacity estimate or a monetary budget. Single-process execution is required by the current task recovery design.

11 Prototype verification and current evidence

The automated suite was rerun for this report and recorded in evidence/automated-tests.xml. All nine test cases passed. The suite uses isolated temporary databases and controlled HTTP responses, allowing deterministic inspection of state changes and provider contracts. Test responses are not reachable as a fallback in the normal application.

Test area	Observed result	What remains unproven
Generation and review lifecycle	Draft, approval, answer visibility and revision checks pass	Pedagogical correctness of a real output
Grounding safeguards	Invented source IDs rejected; insufficient output not published	Entailment by valid cited passages
Index and request consistency	Changed embedding configuration and conflicting requests blocked	Real retrieval relevance and vendor stability
Failure handling	Upstream errors redacted; call cap and origin guard checked	Real platform behaviour under load
Evidence export	Source links and configuration retained without test credentials	Research quality of later saved runs

The local browser verification recorded previously includes creation of the RAG demonstration course, import of intro_to_rag.md as one page and one chunk, and a clear missing-configuration message when indexing was requested. This shows basic interface operation; a one-chunk example cannot evaluate retrieval ranking. No synthetic model output is presented as an actual generated course asset.

A fresh configuration preflight found both text and embedding capabilities unconfigured. The live smoke procedure is therefore prepared but has not been executed. After configuration, it will create a separate demonstration course and run indexing, an English lesson, a quiz and an assignment through the real adapters. It saves task outcomes and evidence bundles, leaves content unapproved, and requires manual inspection.

CA1 readiness is consequently partial at the system level: the literature and design manuscripts are available, and the software path has controlled integration evidence, but real core text generation with RAG still needs demonstration. This is a specific dependency on provider configuration and verification rather than a result that can be inferred from passing unit or integration tests.

12 Evaluation plan and acceptance procedure

The immediate acceptance exercise uses permitted course material to demonstrate ingestion, retrieval, text generation, provenance and teacher review. At least two distinct source sections should be used to show meaningful selection rather than merely returning the only available chunk. Demonstrate both an explanation and assessment items, then test an objective that the supplied material cannot support.

For each live case, retain the source hash, request, retrieved chunk identifiers and text, model configuration, prompt version, output, task status and elapsed time. Annotate incorrect statements, unsupported answers, unsuitable difficulty and ambiguous questions. A successful HTTP response or a schema-valid object alone is not sufficient for accepting a case as usable teaching material.

Controlled study after the live smoke test

The planned comparison uses one permitted course, 12 development objectives and 24 held-out test objectives, separated by chapter where practical. Conditions are A: no retrieved course context; B: retrieval with difficulty labels; and C: retrieval with explicit cognitive constraints. All conditions use the same model, output schema and task settings. The current production path implements the C-style workflow; dedicated A/B experiment conditions still need implementation.

The proposed full run contains 24 objectives × 3 demand levels × 3 conditions = 216 one-question assets. A smaller pilot will first estimate annotation effort. These numbers are planning values, not collected samples or a power calculation. If question format or language changes, results should be stratified rather than silently pooled.

A course-competent rater assesses correctness, support, answerability and intended demand without seeing condition labels. Where possible, a second rater independently scores a subset and disagreements are recorded before adjudication. Retrieval support recall is computed against annotated relevant chunks; absent gold support is reported as not assessable, rather than converted to zero.

Report failures and abstentions over every request, as well as quality conditional on a valid output. Compare matched objectives and use objective-level resampling for uncertainty estimates. Keep teacher-edited versions separate. Without student response data and an appropriate study design, conclusions concern material quality and authoring usability, not learning gains or calibrated item difficulty.

13 Remaining work and milestone plan

Next action	Completion evidence	Dependency
Configure real text and embeddings	Live indexing and three saved text cases	Provider addresses, models and local keys
Review course-grounded outputs	Source-supported explanation and assessment demonstration	Permitted material and course expertise
Freeze CA1 package	Report, code snapshot, evidence and demonstration script agree	Supervisor format and scope feedback
Prepare controlled comparisons	A/B/C conditions and blinded rating records	Stable generator and a pilot dataset
Continue CA2 integration	Reviewed speech and visuals linked to text versions	Real media API tests

Before October, priority is to finish a small but genuine CA1 demonstration and preserve its evidence. The report and review can be discussed with the supervisor now; they should be updated after the first live cases. Formal submission dates remain subject to the course schedule. No unverified date or minimum report length is imposed by this draft.

The most consequential current risks are unavailable course material, incompatible API responses, unsupported generated claims and limited evaluation time. The response is to keep the first course and question formats bounded, record failed runs, and reserve time for manual verification. More model calls or a more elaborate interface do not by themselves resolve these risks.

This milestone establishes a reviewable technical and academic foundation. Final CA1 completion requires replacing the current live-validation gap with actual provider evidence and human review. The literature, design and test artefacts prepared here are intended to make that remaining work concrete and easy to verify.

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